
D-BOX COMPATIBILITY GUIDE

A document describing concepts and best practices for D-BOX Software Compatibility

Live Motion SDK

Integrating this SDK allows D-BOX to create a customized software module that adds hyper-realistic full-body haptic feedback to the gaming experience. This is computed in real time based on game physics and events. The resulting haptic effects are perfectly synchronized with onscreen action, creating a realistic and immersive experience.

Where to retrieve D-BOX events in your game engine?

D-BOX will recreate what the in-game actor would feel. It is very similar to audio programming. Here is what to look for:

- Physics information in Actor **local** coordinates;
- Audio events.

Haptic feedback is like audio feedback; spatialization, filtering and mixing to create layers.

Telemetry Data Types

FRAME_UPDATE: Data sent at every frame - *RPM, Local Accelerations, etc.*

CONFIG_UPDATE: Data sent when a configuration change occurs - *New vehicle, new weapon, etc.*

ACTION: Impulse that is sent when an action occurs - *Start vehicle engine, menu selection, etc.*

ACTION_WITH_DATA: Impulse that is sent with parameters - *Explosion considering distance from player, orientation, etc.*

Data centralization

D-BOX events are all centralized within one enumerated type (enum). Data is regrouped in a contextual structure (struct) and sent as a package with the event.

Live Motion SDK Methods Cycle

```

////////////////////////////////////
// Start application
dbox::LiveMotion::Initialize(APP_KEY, APP_BUILD);

// Register D-BOX events from the enum and their corresponding data struct.
dbox::LiveMotion::RegisterEvent(PLAYER_TELEMETRY, dbox::EM_FRAME_UPDATE,
PlayerTelemetry::GetStructInfo());

// Open Motion Output Device
dbox::LiveMotion::Open();

// Un-mute the D-BOX System and fade-in Motion Stream
dbox::LiveMotion::Start();

// Game Loop, send events to D-BOX when they occur, at every frame or on impulse
// Pass event and corresponding data struct with updated data
dbox::LiveMotion::PostEvent(PLAYER_TELEMETRY, oPlayerTelemetry);

// Mute the system and fade-out remaining Motion data, user paused or is quitting
dbox::LiveMotion::Stop();

// Close Motion Output Device
dbox::LiveMotion::Close();

// End application
dbox::LiveMotion::Terminate();
////////////////////////////////////

```

These methods must **ALWAYS** be paired:

Initialize	Terminate
Open	Close
Start	Stop

See Sample files for detailed integration and notes.

See D-BOX Live Motion SDK Header files for descriptors and a list of available fields.

Less Is More

D-BOX has been creating full-body haptic feedback for over a decade and knows what makes an experience stand out. The more information we have when something happens, the more we can create specific dynamics and immersion. More information does not mean more events. If too many things are happening at once, coherence and enjoyability can be reduced greatly. It is better to have 10 really good effects that happen often opposite to 100 effects that climb over each other and create a big random rumble in the end.

Degrees of Freedom

D-BOX Motion Systems can move in any degree of freedom depending on the type of platform and actuators placement. Typically, a Motion System has freedom in roll, pitch and heave.

Roll: rotation around longitudinal axis;

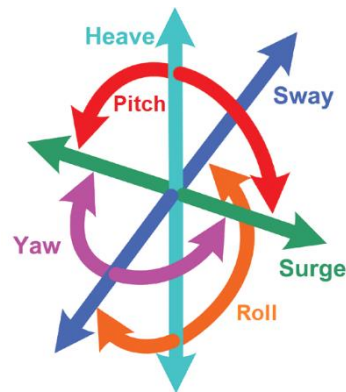
Pitch: rotation around lateral axis;

Heave: translation along vertical axis;

Yaw: rotation around vertical axis;

Sway: translation along lateral axis;

Surge: translation along longitudinal axis.



FAQ

Q: Is the SDK free?

A: Yes. However, there are fees associated with the haptics design portion done by D-BOX.

Q: How can I try the resulting D-BOX integration and Motion Code?

A: you need access to a D-BOX Motion System.

Q: I can't find a descriptor field for what I want to send as telemetry?

A: You can use the CUSTOM frame and configuration fields.

Q: What are the SDKs system requirements?

A: The SDK is compiled in C++ on Windows - 32- or 64-bit libraries. Compatible with Visual Studio 2010, 2015 and 2017.

Q: Will end users be able to see the events sent to D-BOX?

A: The information output towards the D-BOX Motion System is encrypted and can only be read by the D-BOX Motion Code, which is custom created for your specific content.